a

Signature replication in external cohorts

GSE162694 | GSE135251

Fibrosis
- GSE162694: Genome-wide: ρ = 0.42, 64.3% same direction (n = 18,493); Shared DEGs: ρ = 0.67, 97.8% same direction (n = 46)
- GSE135251: Genome-wide: ρ = 0.34, 62.6% same direction (n = 16,160); Shared DEGs: ρ = 0.74, 98.0% same direction (n = 49)

Steatosis
- GSE162694: Genome-wide: ρ = 0.57, 69.0% same direction (n = 16,313); Shared DEGs: ρ = 0.88, 99.4% same direction (n = 683)
- GSE135251: Genome-wide: ρ = 0.49, 65.8% same direction (n = 15,094); Shared DEGs: ρ = 0.88, 98.4% same direction (n = 853)

Inflammation
- GSE162694: Genome-wide: ρ = 0.53, 67.8% same direction (n = 17,175); Shared DEGs: ρ = 0.87, 99.1% same direction (n = 761)
- GSE135251: Genome-wide: ρ = 0.48, 65.4% same direction (n = 15,560); Shared DEGs: ρ = 0.89, 98.3% same direction (n = 969)

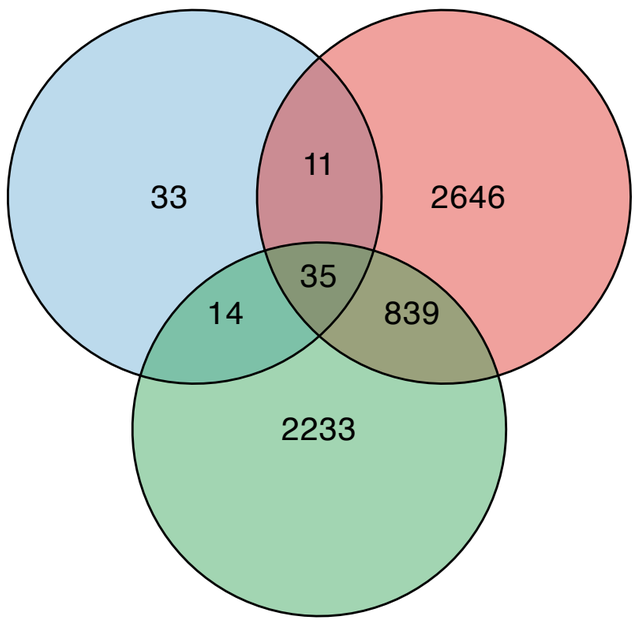
$\log_2$FC — external

$\log_2$FC — discovery

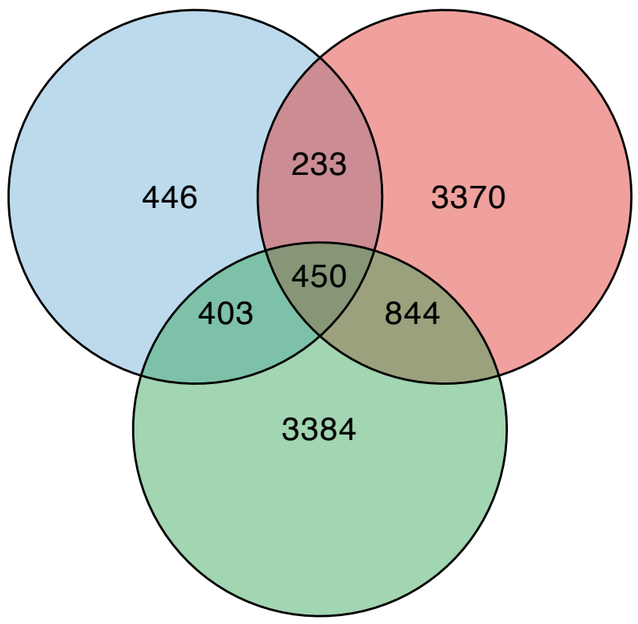
Gene status Not significant | Validation-only | Discovery-only | Discordant direction | Replicated (concordant)

b

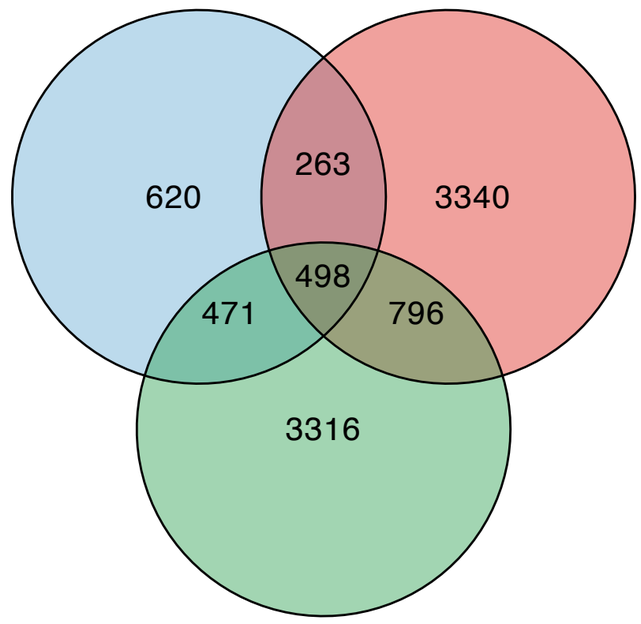
Fibrosis
Discovery fibrosis | GSE162694



Steatosis (discovery) vs NAS (external)
Discovery steatosis | GSE162694



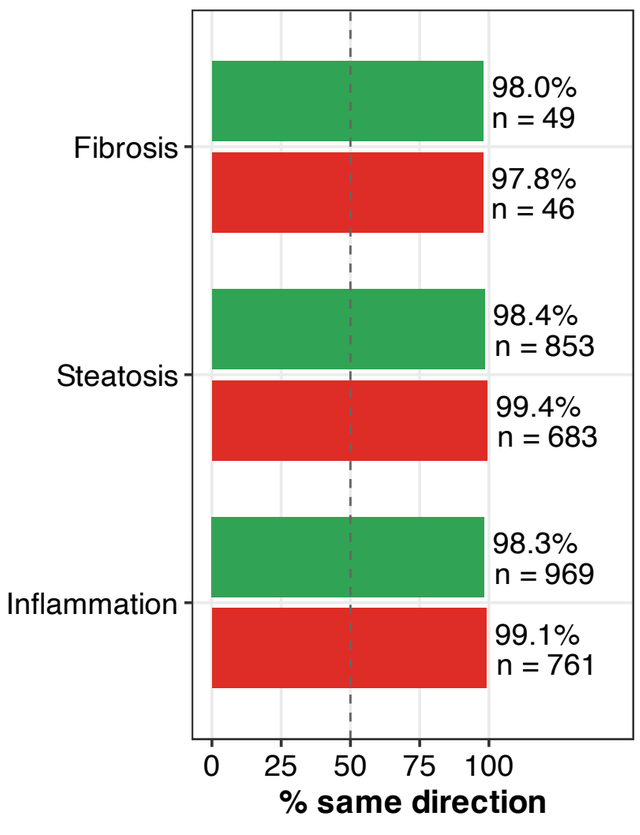
Inflammation (discovery) vs NAS (external)
Discovery inflammation | GSE162694



c

GSE135251

Directional concordance



GSE135251

Effect-size correlation



GSE162694 | GSE135251

d

GSE135251

Replicated KEGG pathways (fibrosis)

FDR ≤ 0.10, ≥ 3 genes; discovery and ≥ 1 external cohort

